

INTEGRATION PRACTICE

Evaluate the integrals below by **first** identifying the **technique** that is appropriate. Proceed only far enough to be sure your strategy will work. Then work the problems according to **which one is hardest for you**.

trig id.

$$1. \int_0^{\pi/4} \sin^2(2\theta) dx = \frac{1}{2} \int_0^{\pi/4} (1 - \cos(4\theta)) d\theta = \frac{1}{2} \left(\theta - \frac{1}{4} \sin(4\theta) \right) \Big|_0^{\pi/4}$$

$$= \frac{1}{2} \left(\left[\frac{\pi}{4} - \frac{1}{4} \sin(\pi) \right] - \left(0 - \frac{1}{4} \sin(0) \right) \right)$$

$$= \frac{\pi}{8}$$

u-sub

$$2. \int x \sqrt{x^2 + 1} dx = \frac{1}{2} \int u^{1/2} du = \frac{1}{2} \cdot \frac{2}{3} u^{3/2} + C = \frac{1}{3} (x^2 + 1)^{3/2} + C$$

$$\text{let } u = x^2 + 1$$

$$du = 2x dx$$

$$\frac{1}{2} du = x dx$$

trig sub

$$3. \int \frac{\sqrt{x^2 + 4}}{x} dx = \int \frac{2 \tan \theta \cdot 2 \sec \theta \tan \theta d\theta}{2 \sec \theta} = 2 \int \tan^2 \theta d\theta = 2 \int (\sec^2 \theta - 1) d\theta$$

$$\text{let } x = 2 \sec \theta$$

$$dx = 2 \sec \theta \tan \theta d\theta$$

$$\sqrt{x^2 + 4} = 2 \tan \theta$$

$$\text{so } \frac{\sqrt{x^2 + 4}}{2} = \tan \theta;$$

$$\left\| = 2(\tan \theta - \theta) + C = 2 \left(\frac{\sqrt{x^2 + 4}}{2} - \arctan \left(\frac{\sqrt{x^2 + 4}}{2} \right) \right) + C$$

$$4. \int_2^6 \ln(t) dt = \left[t \ln(2) \right]_2^6 - \int_2^6 t \cdot \frac{1}{t} dt = 6 \ln(6) - 2 \ln(2) - \int_2^6 1 dt$$

$$u = \ln(t) \quad dv = dt$$

$$du = \frac{1}{t} dt \quad v = t$$

$$= 6 \ln(6) - 2 \ln(2) - 4$$

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$$5. \int \frac{2x-3}{x^2-3x-4} dx = \int \frac{du}{u} = \ln|u| + C = \ln|x^2-3x-4| + C$$

$$\text{let } u = x^2 - 3x - 4$$

$$du = (2x-3) dx$$

u-sub

$$6. \int \frac{6x}{x^2-3x-4} dx = \int \frac{6x dx}{(x-4)(x+1)} = \int \left(\frac{24/5}{x-4} + \frac{6/5}{x+1} \right) dx$$

$$\frac{6x}{x^2-3x-4} = \frac{A}{x-4} + \frac{B}{x+1} \quad \parallel \quad = \frac{24}{5} \ln|x-4| + \frac{6}{5} \ln|x+1| + C$$

$$6x = A(x+1) + B(x-4)$$

$$x=4: 24 = 5A, A = 24/5$$

$$x=-1: -6 = B(-5), B = 6/5$$

partial fractions

$$7. \int \sin^6(x) \cos^3(x) dx = \int \sin^6(x) \cos^2(x) \cos x dx \quad \begin{array}{l} \text{let } u = \sin x \\ du = \cos x dx \end{array}$$

$$= \int \sin^6(x) (1 - \sin^2(x)) \cos x dx = \int u^6 (1 - u^2) du$$

$$= \int (u^6 - u^8) du = \frac{1}{7} u^7 - \frac{1}{9} u^9 + C = \frac{1}{7} \sin^7 x - \frac{1}{9} \sin^9 x + C$$

trig id.

$$8. \int_0^{\pi/2} x \sin(7x) dx = -\frac{x}{7} \cos(7x) \Big|_0^{\pi/2} + \frac{1}{7} \int_0^{\pi/2} \cos(7x) dx$$

$$u=x \quad dv=\sin(7x) dx \quad \parallel \quad \begin{array}{l} du=dx \\ v=-\frac{1}{7} \cos(7x) \end{array} \quad \parallel \quad = \left(-\frac{\pi}{14} \cos\left(\frac{\pi}{2}\right) + 0 \cdot \cos(0) \right) + \left[\frac{1}{49} (\sin(7x)) \right]_0^{\pi/2}$$

$$= 0 + \left(\frac{1}{49} \right) \left[\sin\left(\frac{7\pi}{2}\right) - \sin 0 \right] = \frac{-1}{49}$$

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